A METHOD FOR SECURE COMMUNICATION VIA RECEIVER-GENERATED KEY SEEDS

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Title: A METHOD FOR SECURE COMMUNICATION VIA RECEIVER-GENERATED KEY SEEDS

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Cross-Reference to Related Applications

This application builds upon concepts disclosed in U.S. Patent Application No. 19/640,945, titled "A METHOD FOR ASYNCHRONOUS SECURE COMMUNICATION WITH POST-HOC KEY DERIVATION," and U.S. Patent Application No. 19/561,259, titled "Protocol for Encoding Map Rotation for Forward Secrecy." The contents of these applications are hereby incorporated by reference in their entirety.

Abstract

A method for secure communication wherein the keying material used by a sender to encrypt a message is first generated by the intended receiver and transmitted to the sender. This inversion of the traditional key exchange paradigm makes the sender-to-receiver (downlink) channel as cryptographically strong as the receiver-to-sender (uplink) channel. The receiver generates a set of key seeds and transmits them to the sender. The sender uses these seeds to

generate a one-time-use encryption key, encrypts the message, and transmits the encrypted message to the receiver. The receiver, having possession of the original seeds, can independently generate the same key to decrypt the message. Because the final encryption key is never transmitted, the method provides a foundation for a new class of secure session protocols to replace standards like TLS/SSL that are vulnerable to man-in-the-middle attacks during the initial key exchange.

Background of the Invention

In nearly all communication systems, the downlink (from a central entity to a user) is inherently more vulnerable to interception than the uplink (from the user to the central entity). An attacker can monitor the communications to a single receiver and potentially intercept messages from all senders communicating with that receiver. This vulnerability is most acute during the key exchange phase of traditional cryptographic protocols.

Prevailing standards for establishing secure sessions, such as Transport Layer Security (TLS) and its predecessor, Secure Sockets Layer (SSL), operate on a "key-first" paradigm. A client and server perform a handshake procedure to negotiate cryptographic parameters and exchange or collaboratively generate a session key. Only after this key is established is the application data transmitted. The entire security of the session depends on the security of this initial handshake. An attacker who can intercept this key exchange in real-time (a "Man-in-the-Middle" or MITM attack) can compromise the entire session that follows. While U.S. Patent Application No. 19/640,945 improves security by sending the message before the final key, it does not fundamentally alter the sender-originating key model. There is a need for a method that eliminates the transmission of the message encryption key entirely, thereby creating a new foundation for secure session protocols immune to these handshake vulnerabilities.

Summary of the Invention

The present invention is a method for secure communication that completely inverts the flow of keying material, thereby making the sender-to-receiver channel as difficult to compromise as the receiver-to-sender channel. The core principle is that the receiver, not the sender, generates the initial seeds from which the message encryption key is derived. This method provides a novel framework for establishing secure sessions that can serve as a direct replacement for the vulnerable handshake mechanisms of TLS and SSL.

The method begins with the entity that will be the receiver of a message. The receiver generates one or more sets of cryptographic seeds. These seeds are then transmitted over the receiver's uplink channel to the sender. At a later time, when the sender has a message to transmit, it uses one of the received seed-sets to generate a one-time-use encryption key. The message is encrypted with this key and transmitted to the receiver. The receiver, which already possesses the original seed-set, performs the same key generation process to derive the correct key and decrypt the message.

This method ensures that the key used to encrypt the critical message is never transmitted. This fundamental shift away from the "key-first" paradigm of TLS/SSL neutralizes the threat of MITM attacks during session initialization. By having the client (receiver) provide the server (sender) with a queue of single-use key seeds in advance, a secure session can be established without transmitting any secrets at the time of connection, making the entire class of real-time handshake interception attacks obsolete.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a flowchart illustrating the complete lifecycle of the secure communication method, showing the distinct roles of the Future Receiver and Future Sender.

FIG. 2 is a conceptual diagram contrasting the data flow of a traditional key exchange with the receiver-generated key seed method of the present invention.

Detailed Description of the Invention

Referring to **FIG. 1**, the lifecycle of the communication method is shown. The process involves two roles: a Future Receiver (101) and a Future Sender (102).

The process begins with the Future Receiver (101). In step (110), the Future Receiver generates one or more sets of cryptographic key seeds. These seeds are the raw material from which the final encryption key will be derived. In step (112), the Future Receiver transmits these seeds to the Future Sender over its uplink channel. This transmission may optionally be encrypted. The Future Sender receives and stores these seeds for later use (150).

At some arbitrary time in the future, the roles shift. The Future Sender (102) now has a message M to transmit to the now-Receiver (101). In step (152), the Sender retrieves a set of the previously received seeds. In step (154), the Sender uses a deterministic key generation algorithm (e.g., a standard Key Derivation Function or KDF) to generate an encryption key, K-msg, from the seeds.

In step (156), the Sender encrypts the message M using K-msg to produce the encrypted message. This encrypted message is then transmitted over the Receiver's downlink to the Receiver (158).

The Receiver (101) receives the encrypted message. In step (114), the Receiver, who has the original seeds from step (110), applies the identical key generation algorithm to independently produce either the same encryption key, or another viable encryption key (K-msg). Finally, in step (116), the Receiver uses this self-generated key to decrypt the message and recover the original plaintext. At no point was K-msg ever transmitted.

Referring to **FIG. 2**, the paradigm shift of the present invention is contrasted with a traditional key exchange. In a traditional system (200), such as the handshake in TLS/SSL, the Sender (201) and Receiver (202) exchange keying material over their respective channels, creating a window of opportunity for a Man-in-the-Middle (MITM) attacker (204) to intercept a key on the downlink (203) and compromise the session.

In the present invention (250), the flow is reversed. The Receiver (252) generates the key seeds and transmits them over its uplink (253) to the Sender (251). The Sender uses these

seeds to encrypt the message, which is then sent over the downlink (254). The MITM attacker (255) now gains no advantage by monitoring the downlink, as the key is never sent. The security of the downlink is now contingent on the security of the much less vulnerable uplink.

Claims

What is claimed is:

1. A method for secure communication between a first party and a second party, comprising the steps of:
    - at the first party, generating a set of cryptographic seeds;
    - transmitting said seeds from the first party to the second party;
    - at the second party, generating a message encryption key from the received seeds using a deterministic algorithm;
    - at the second party, encrypting a message using said message encryption key;
    - transmitting the encrypted message from the second party to the first party;
    - at the first party, independently generating the same message encryption key or an alternate-viable encryption key from the original seeds using the same deterministic algorithm; and
    - at the first party, decrypting the received message using the independently generated key.
2. The method of claim 1, wherein the cryptographic seeds are transmitted from the first party to the second party over a first communication channel, and the encrypted message is transmitted from the second party to the first party over a second, different communication channel.
3. The method of claim 1, wherein the transmission of the seeds is encrypted using a key separate from the message encryption key.
4. The method of claim 1, wherein the first party generates a plurality of sets of cryptographic seeds and transmits them to the second party for storage in a queue.
5. The method of claim 4, further comprising the step of the second party selecting one set of seeds from the plurality based on a shared context known to both parties, and using only the selected set to generate the message encryption key.

6. The method of claim 5, wherein the shared context is an index into a list of seed-sets, and wherein a seed-set is discarded by both parties after use.

7. The method of claim 1, wherein the deterministic algorithm is a Key Derivation Function (KDF).

8. The method of claim 1, applied as a replacement for the handshake and key exchange mechanism in a secure session protocol, wherein the first party is a client and the second party is a server.

9. The method of claim 8, wherein the client transmits a plurality of seed-sets to the server in advance of a secure session, and the server consumes one seed-set from the plurality to encrypt session data for transmission to the client.

Drawings

FIG. 1: Flowchart of the secure communication method.

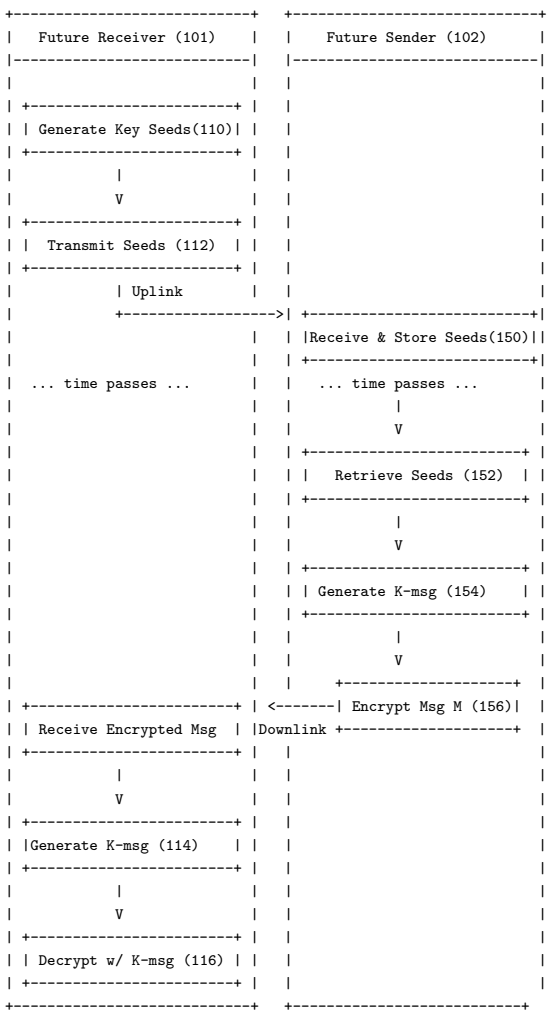


FIG. 2: Conceptual diagram contrasting methods.

